Event Study in Accounting & Finance Analytics

© Mohammad M. Rahaman, PhD
Canada Research Chair (CRC) in International Finance & Competitiveness
Sobey School of Business
Saint Mary's University

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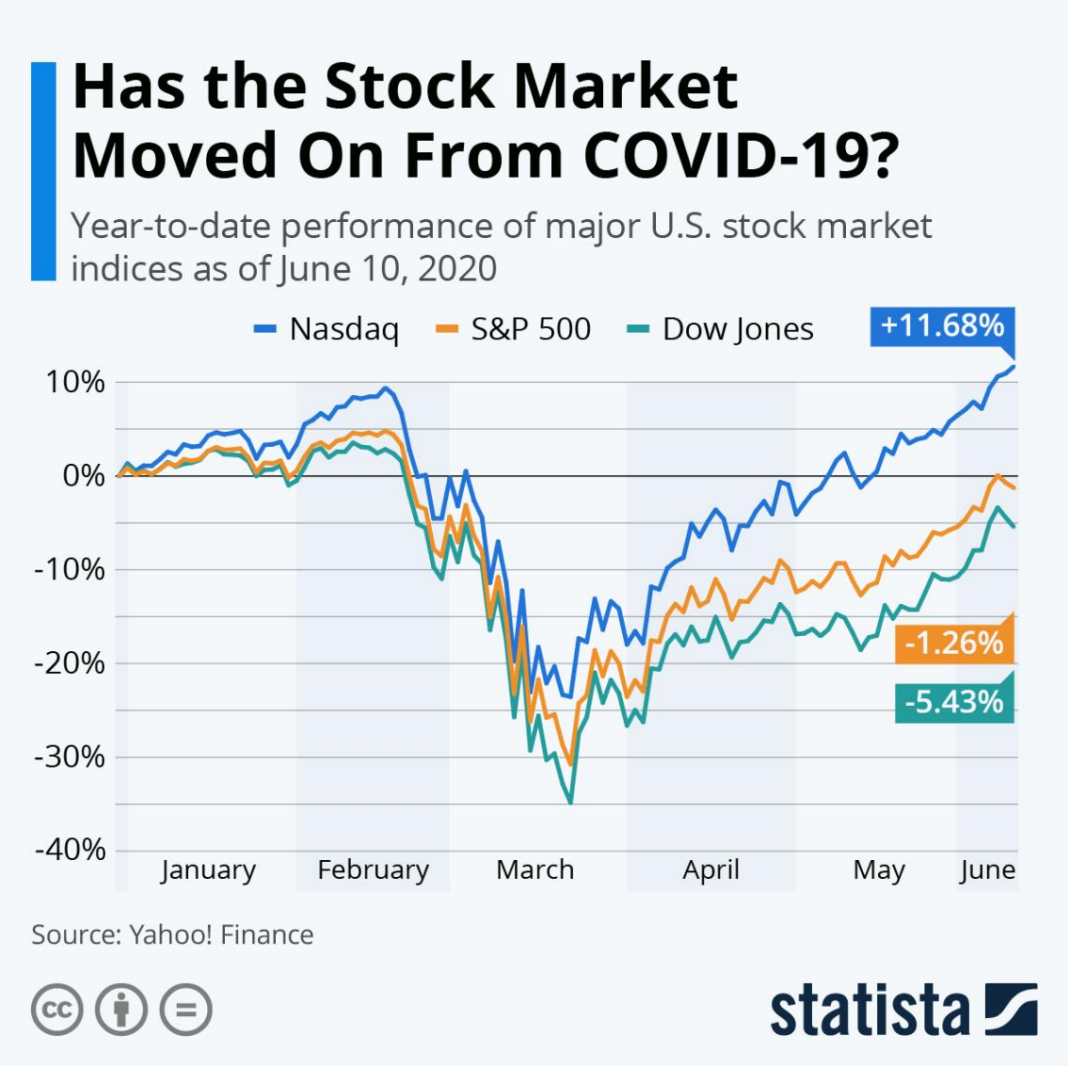
Saint Mary's University

COVID-19 & the Stock Market



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US-China Trade War 2018-2019

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Buyout Announcement

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Capital Market Reaction to News

- Event studies examine prices and returns over time (particularly around the arrival of new information.)
- Test for evidence of:
  - Under-reaction
  - Over-reaction
  - Early-reaction
  - Delayed-reaction around the event.

- If market is "semi-strong-form efficient", the effects of an event will be reflected immediately in security prices. Thus, a measure of the event's economic impact can be constructed using security prices observed over a relatively short time period.

- Some examples of events include:
  - Mergers and acquisitions.
  - Earnings announcements .
  - Issues of new debt or equity.
  - Stock splits.
  - Announcements of macroeconomic variables such as the trade figures.
  - Health issue of the key person of a company.

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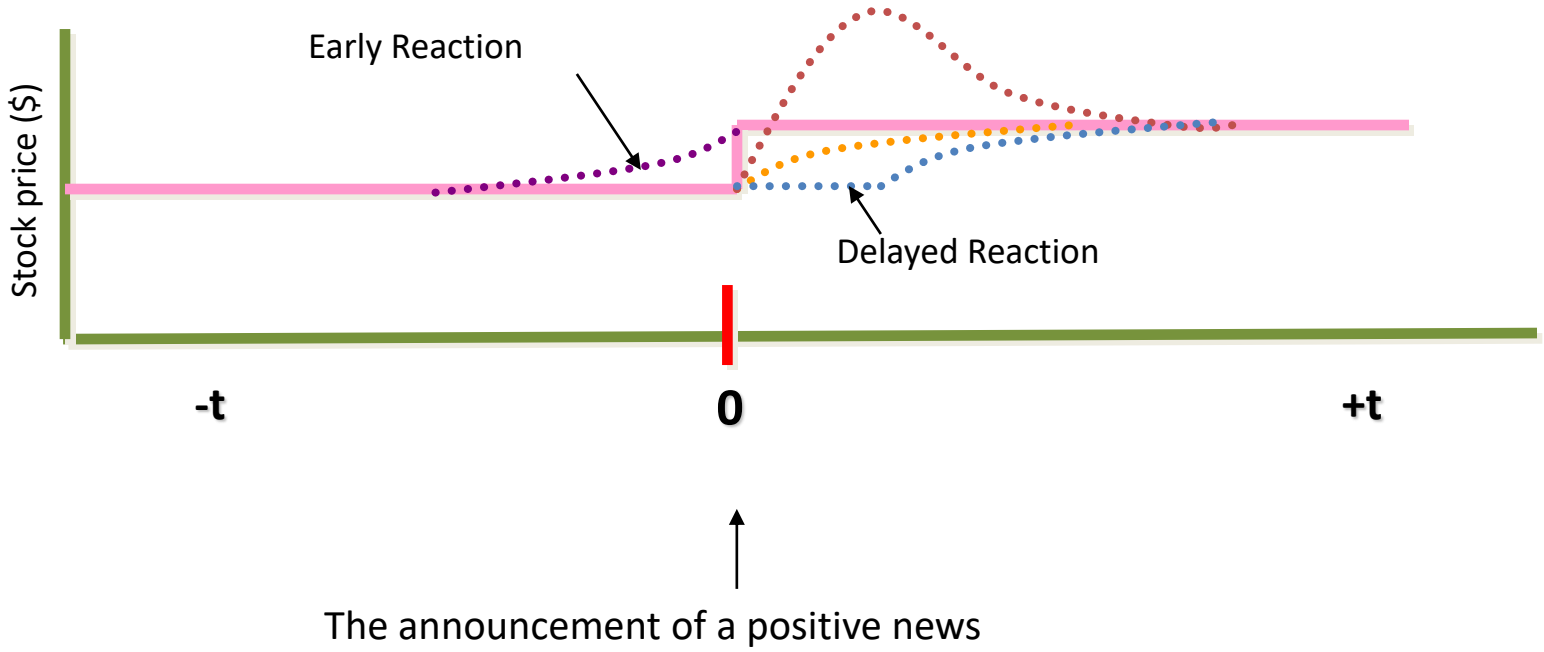
Event Study: Reaction to News



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Event Study Implementation Steps

- Sample Security – Stocks.
- Identify the event of interest (e.g., stock splits, dividend announcement, merger announcement).
- Define the event period (e.g., -10 to +10 days relative to the event date)
- Select the sample (e.g., firms which have in common the incidence of the event of interest).
- Measure the impact by means of abnormal return.
- Estimate the parameters needed to calculate expected returns.
- Calculate cumulative abnormal returns.

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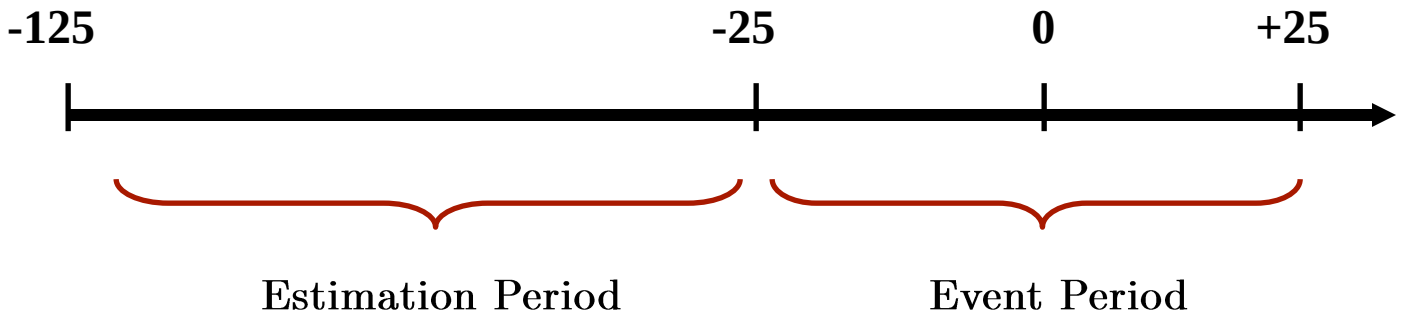
Event Study:
Steps in Implementing an Event

- The event period should start before you think the event has an effect on the stock price. As an example, for merger announcements, a typical choice is from 25 trading days before the announcement day to 25 trading days after the announcement day.

- The estimation period should be a period right before the event period. For merger announcements, a typical choice is 100 trading days before the start of the event period.



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Event Study Implementation Steps

- Abnormal return = Actual realized return – Expected return
  - An example of expected return is the simple market model as follows: $E(R_j|R_{M,t}) = \alpha_0 + \beta_j \times R_{M,t}$
  - Note that the return on security j is conditional on the return on market
  - Once you the expected return and the realized return (from the financial market), the abnormal return is calculated as follows: $AR_{j,t} = R_{j,t} - E(R_j|R_{M,t})$
- Cumulative abnormal return:
  - $CAR_{j,t} = \sum_{-t}^{+t} AR_{j,t}$
  - Aggregate abnormal returns from –t to +t)
- Average cumulative abnormal return over a sample of securities:
  - $CAAR_t = \frac{1}{J}\sum(CAR_{j,t})$, where J = no. of securities in the sample.
- Plot the graph, examine the pattern. Of course, perform hypothesis testing as well.

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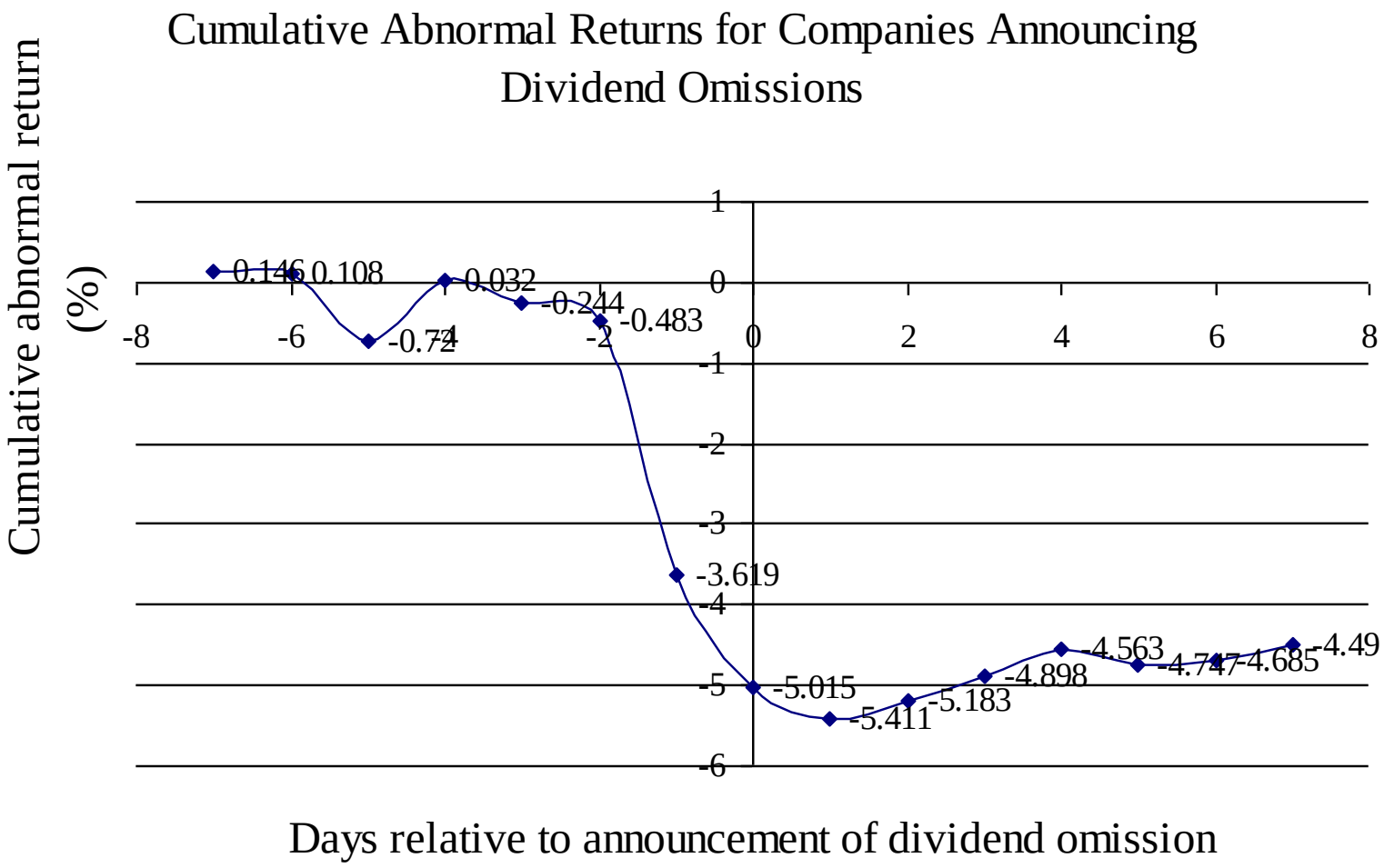
Event Study Example: Dividend Cut



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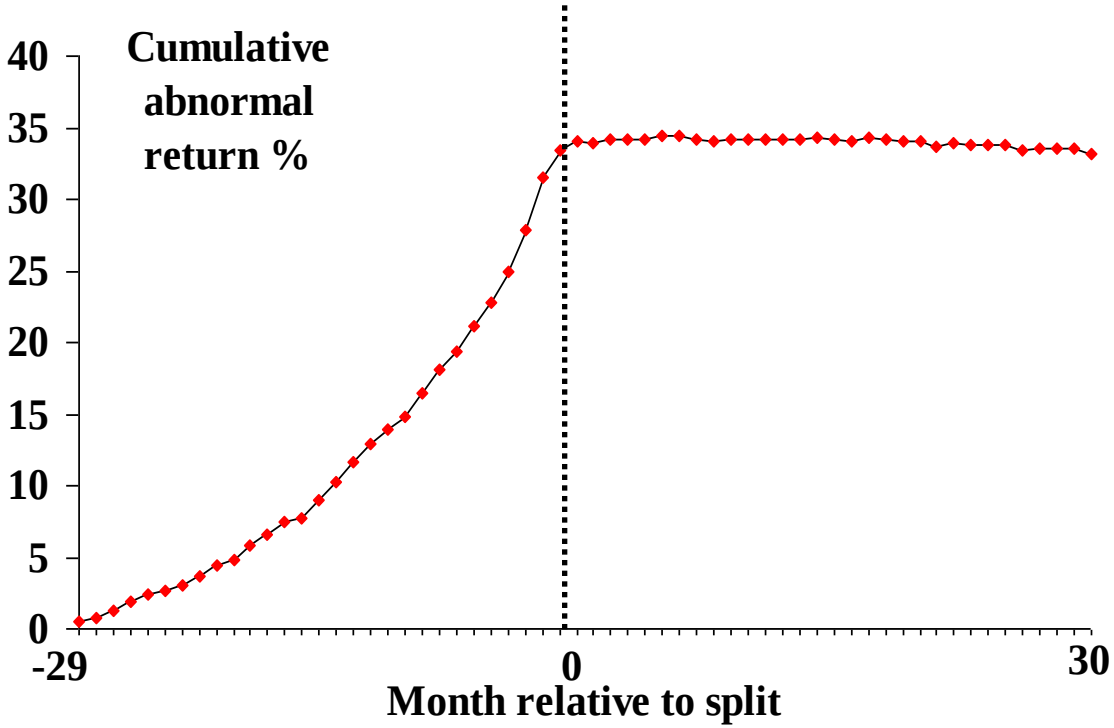
Event Study Example: Stock Split



How stock splits affect value?

Source: Fama, Fisher, Jensen & Roll (1969)

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Event Study: Methodological Assumptions

- The event study methodology assumes that capital markets accurately reflect the economic implications that the analyzed event has for the firm in question, i.e., event studies provide a direct test of market efficiency.
- Events across the sample are independent of one another, i.e., events cannot cluster within time and industry.
- What ever asset pricing model that we are using to calculate the expected return is indeed the right asset pricing model.

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Event Study: Statistical Significance Test

- The main hypotheses for an event study is: $H_0: CAAR = 0$
  - The t-statistics for the test is given by: $t_{CAAR} = \left(\frac{CAAR}{S_{CAAR}}\right) \times \sqrt{N}$, where N is the number of securities.
  - And $S^2_{CAAR} = \frac{1}{N-1}\sum_{i=1}^{N}(CAR_i - CAAR)^2$

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Event Study: Summary

- Event study methodology has been applied to a large number of events including:
  - Dividend increases and decreases.
  - Earnings announcements.
  - Mergers & Acquisitions.
  - Capital Spending.
  - New Issues of Stock.

- The studies generally support the view that the market is respond to information efficiently.

- In fact, the studies suggest that markets may even have some foresight into the future—in other words, news tends to leak out in advance of public announcements.

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